

STA 2002

Lecture 21

Two-way ANOVA

The setup

We are interested in the effects of **sex** and **dietary group** on **systolic blood pressure (SBP)**

Table 12.14 Mean SBP by dietary group and sex

Dietary group		Sex: Male	Sex: Female
SV strict vegetarians	Mean	109.9 (112.3, 99.7, 103.2, ...)	102.6 (105.8, 110.2, 88.5, ...)
LV lactovegetarians	Mean	115.5 (103.0, 126.9, 108.3, ...)	105.2 (101.2, 101.0, 99.4, ...)
NOR normals	Mean	128.3 (132.8, 111.6, 125.5, ...)	119.6 (104.9, 125.6, 133.5, ...)

i *j* *k*

Cell Means Model

$X_{ijk} = \mu_{ij} + \varepsilon_{ijk}, \qquad i = 1, \ldots a;\ j = 1, \ldots b;\ k = 1, \ldots c$

$\mu_{ij} -$ parameters

$\varepsilon_{ijk} \sim i.i.d.N(0, \sigma^2)$

$E(X_{ijk}) = \mu_{ij}$
$Var(X_{ijk}) = Var(\varepsilon_{ijk}) = \sigma^2$

How do we estimate μ_{ij} ?

Use the sample mean $\bar{X}_{ij.} = \frac{1}{c}\sum_{k=1}^{c} X_{ijk}$

Cell Means

	Male	Female	Row Average
SV	μ_{11}	μ_{12}	$\mu_{1\cdot}$
LV	μ_{21}	μ_{22}	$\mu_{2\cdot}$
NOR	μ_{31}	μ_{32}	$\mu_{3\cdot}$
Column Average	$\mu_{\cdot 1}$	$\mu_{\cdot 2}$	$\mu_{\cdot\cdot}$

Row mean:

$$\mu_{i\cdot} = \frac{\sum_{j=1}^{b} \mu_{ij}}{b}$$

Column mean:

$$\mu_{\cdot j} = \frac{\sum_{i=1}^{a} \mu_{ij}}{a}$$

Overall mean:

$$\mu_{\cdot\cdot} = \frac{\sum_{i=1}^{a}\sum_{j=1}^{b} \mu_{ij}}{ab} = \frac{\sum_{i=1}^{a} \mu_{i\cdot}}{a} = \frac{\sum_{j=1}^{b} \mu_{\cdot j}}{b}$$

Row/Column effects

	Male	Female	Row Average
SV	μ_{11}	μ_{12}	$\mu_{1.}$
LV	μ_{21}	μ_{22}	$\mu_{2.}$
NOR	μ_{31}	μ_{32}	$\mu_{3.}$
Column Average	$\mu_{.1}$	$\mu_{.2}$	$\mu_{..}$

- What's the effect of SV? $\alpha_1 = \mu_{1.} - \mu_{..}$
- What's the effect of Female? $\beta_2 = \mu_{.2} - \mu_{..}$
- Note that $\mu_{12} = \mu_{..} + (\mu_{1.} - \mu_{..}) + (\mu_{.2} - \mu_{..}) + (\mu_{12} - \mu_{1.} - \mu_{.2} + \mu_{..})$
- Another expression for μ_{12} is

$$\mu_{12} = \mu_{..} + \alpha_1 + \beta_2 + \gamma_{12}$$

where $\gamma_{12} = \mu_{12} - \mu_{1.} - \mu_{.2} + \mu_{..}$ is called an interaction term

Main Effects Model

The kth observation in the cell of the ith row and j^{th} column

$$X_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ijk}$$

Overall mean
$\mu = \mu_{..}$

Effect of the ith value of the row factor
$\alpha_i = \mu_{i.} - \mu_{..}$

Effect of the jth value of the column factor
$\beta_j = \mu_{.j} - \mu_{..}$

Interaction effect of the ith value of the row factor and the jth value of the column factor $\gamma_{ij} = \mu_{ij} - \mu_{i.} - \mu_{.j} + \mu_{..}$

Error term $\sim N(0, \sigma^2)$

$$\sum \alpha_i = \sum \beta_j = 0, \quad \sum \gamma_{ij} = 0 \quad \text{for all } i \quad \sum \gamma_{ij} = 0 \quad \text{for all } j$$

An **interaction effect** between two variables is defined as one in which the effect of one variable depends on the level of the other variable.

Main Effects Model

		Sex	
Dietary group		Male	Female
SV strict vegetarians	Mean	109.9	102.6
LV lactovegetarians	Mean	115.5	105.2
NOR normals	Mean	128.3	119.6

The kth observation in the cell of the ith row and j^{th} column

$$X_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ijk}$$

Overall mean SBP

Effect of dietary group

Effect of the *sex*

Interaction effect of dietary group and sex

Error term $\sim N(0, \sigma^2)$

Impact of the Interaction

- If there is no interaction, i.e., $X_{ijk} = \mu + \alpha_i + \beta_j + \varepsilon_{ijk}$, it can be shown that

$$\mu_{ij} - \mu_{i'j} = \alpha_i - \alpha_{i'} \text{ and } \mu_{ij} - \mu_{ij'} = \beta_j - \beta_{j'}.$$

You can directly comment on the effect of Diet on SBP, e.g., SV leads to the change of SBP in the amount $\alpha_1 - \alpha_3$ compared to Normal diet, regardless of the gender of the person.

- If there is interaction, i.e., $X_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ijk}$, it can be shown that

$$\mu_{ij} - \mu_{i'j} = \alpha_i - \alpha_{i'} + \gamma_{ij} - \gamma_{i'j} \text{ and } \mu_{ij} - \mu_{ij'} = \beta_j - \beta_{j'} + \gamma_{ij} - \gamma_{ij'}.$$

You can no longer directly comment on the effect of Diet on SBP. Now SV, still compared to Normal diet, leads to the change of SBP in the amount $\alpha_1 - \alpha_3 + \gamma_{11} - \gamma_{31}$ for males and $\alpha_1 - \alpha_3 + \gamma_{12} - \gamma_{32}$ for females.

- The interpretation based on row or column effects alone no longer makes sense in the presence of interaction.

Parameter Estimation

The kth observation in the cell of the ith row and j^{th} column

$$X_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + \varepsilon_{ijk}$$

Overall mean SBP

$\mu = \mu_{..}$

Row effect

$\alpha_i = \mu_{i.} - \mu_{..}$

Column effect

$\beta_j = \mu_{.j} - \mu_{..}$

Interaction effect row and column factors

$\gamma_{ij} = \mu_{ij} - \mu_{i.} - \mu_{.j} + \mu_{..}$

Error term $\sim N(0, \sigma^2)$

We first estimate $\mu_{ij}, \mu_{i.}, \mu_{.j}, \mu_{..}$ and then estimate $\alpha_i, \beta_j, \gamma_{ij}$

Parameter Estimation

cell average $\overline{X}_{ij\cdot} = \frac{1}{c}\sum_{k=1}^{c} X_{ijk},$ estimates μ_{ij}

row average $\overline{X}_{i\cdot\cdot} = \frac{1}{bc}\sum_{j=1}^{b}\sum_{k=1}^{c} X_{ijk},$ estimates $\mu_{i\cdot}$

column average $\overline{X}_{\cdot j\cdot} = \frac{1}{ac}\sum_{i=1}^{a}\sum_{k=1}^{c} X_{ijk},$ estimates $\mu_{j\cdot}$

overall average $\overline{X}_{\cdots} = \frac{1}{abc}\sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{c} X_{ijk},$ estimates $\mu_{\cdot\cdot}$

The error term ε_{ijk} is estimated by $X_{ijk} - \bar{X}_{ij\cdot}$ since $X_{ijk} = \mu_{ij} + \varepsilon_{ijk}$.

The variance decomposition

The deviation of an individual observation from the overall mean

$$X_{ijk} - \bar{X}_{...} = \left(X_{ijk} - \bar{X}_{ij.}\right) + \left(\bar{X}_{i..} - \bar{X}_{...}\right) + \left(\bar{X}_{.j.} - \bar{X}_{...}\right) + \left(\bar{X}_{ij.} - \bar{X}_{i..} - \bar{X}_{.j.} + \bar{X}_{...}\right)$$

error term **row effect** **column effect** **interaction effect**

cell average $\bar{X}_{ij.} = \frac{1}{c}\sum_{k=1}^{c} X_{ijk},$

row average $\bar{X}_{i..} = \frac{1}{bc}\sum_{j=1}^{b}\sum_{k=1}^{c} X_{ijk},$

column average $\bar{X}_{.j.} = \frac{1}{ac}\sum_{i=1}^{a}\sum_{k=1}^{c} X_{ijk},$

overall average $\bar{X}_{...} = \frac{1}{abc}\sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{c} X_{ijk},$

a rows
b columns
c repeats in each cell

The variance decomposition

It can be shown that:

$$\sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{c}(X_{ijk}-\overline{X}_{...})^2 = bc\sum_{i=1}^{a}(\overline{X}_{i..}-\overline{X}_{...})^2 + ac\sum_{j=1}^{b}(\overline{X}_{.j.}-\overline{X}_{...})^2 + c\sum_{i=1}^{a}\sum_{j=1}^{b}(\overline{X}_{ij.}-\overline{X}_{i..}-\overline{X}_{.j.}+\overline{X}_{...})^2 + \sum_{i=1}^{a}\sum_{j=1}^{b}\sum_{k=1}^{c}(X_{ijk}-\overline{X}_{ij.})^2$$

$$SS(TO) = SS(A) + SS(B) + SS(AB) + SS(E)$$

SS(A) is the row sum of squares

SS(B) is the column sum of squares

SS(AB) is the interaction sum of squares

SS(E) is the error sum of squares

a rows
b columns
c repeats in each cell

The test

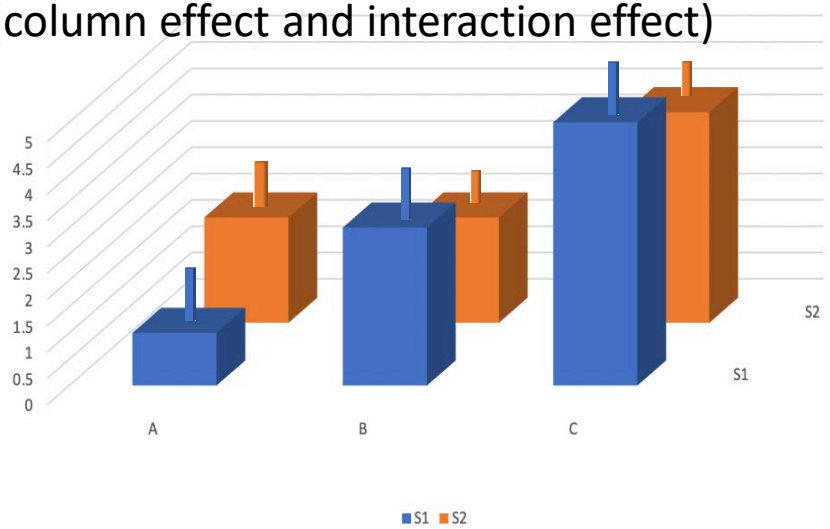
(1) Test for the presence of row effects: H_0: all $\alpha_i = 0$ vs. H_1: at least one $\alpha_i \neq 0$. This is a test for the effect of dietary group on SBP level after controlling for the effect of sex.

(2) Test for the presence of column effects: H_0: all $\beta_j = 0$ vs. H_1: at least one $\beta_j \neq 0$ This is a test for the effect of sex on SBP level after controlling for the effect of dietary group.

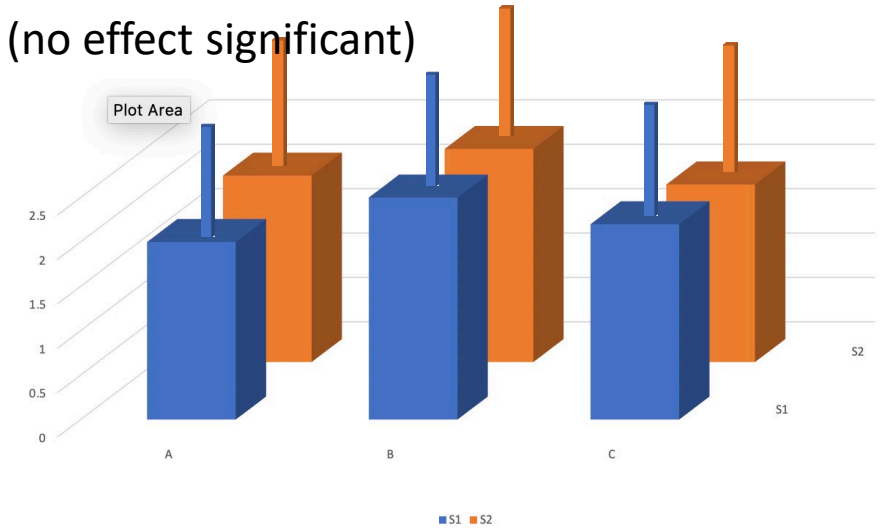
(3) Test for the presence of interaction effects: H_0: all $\gamma_{ij} = 0$ vs. H_1: at least one $\gamma_{ij} \neq 0$. This is a test of whether or not there is a differential effect of dietary group between males and females. For example, dietary group may have an effect on SBP only among men.

Data variation factors into the test results.

(column effect and interaction effect)



(no effect significant)



The test

Table 9.4-4 Two-way ANOVA table, c observations per cell

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F
Factor A (row)	SS(A)	$a-1$	$\text{MS(A)} = \dfrac{\text{SS(A)}}{a-1}$	$\dfrac{\text{MS(A)}}{\text{MS(E)}}$
Factor B (column)	SS(B)	$b-1$	$\text{MS(B)} = \dfrac{\text{SS(B)}}{b-1}$	$\dfrac{\text{MS(B)}}{\text{MS(E)}}$
Factor AB (interaction)	SS(AB)	$(a-1)(b-1)$	$\text{MS(AB)} = \dfrac{\text{SS(AB)}}{(a-1)(b-1)}$	$\dfrac{\text{MS(AB)}}{\text{MS(E)}}$
Error	SS(E)	$ab(c-1)$	$\text{MS(E)} = \dfrac{\text{SS(E)}}{ab(c-1)}$	
Total	SS(TO)	$abc-1$		

The test

The distribution of SS(E):

For each cell $$\frac{\sum_{k=1}^{c}(X_{ijk}-\overline{X}_{ij.})^2}{\sigma^2} \sim \chi^2(c-1)$$

Errors in cells are independent. Thus the sum of them (a total of ab cells):

$$\mathrm{SS(E)}/\sigma^2 \sim \chi^2[ab(c-1)]$$

a rows
b columns
c repeats in each cell

The test

(1) Test for the presence of row effects: H_0: all $\alpha_i = 0$ vs. H_1: at least one $\alpha_i \neq 0$. This is a test for the effect of dietary group on SBP level after controlling for the effect of sex.

The null hypothesis for A: $H_A: \alpha_1 = \alpha_2 = \cdots = \alpha_a = 0$

Under the null:

$$SS(A)/\sigma^2 \sim \chi^2(a-1)$$

$$F_A = \frac{SS(A)/(a-1)}{SS(E)/[ab(c-1)]} \sim F(a\text{-}1,\ ab(c\text{-}1))$$

Table 12.14 **Mean SBP by dietary group and sex**

Dietary group		Sex: Male	Sex: Female
SV strict vegetarians	Mean	109.9	102.6
LV lactovegetarians	Mean	115.5	105.2
NOR normals	Mean	128.3	119.6

SOURCE	DF	SS	F VALUE	PR > F
STUDY	2	51806.42069945	132.24	0.0001
ERROR	745	145934.76850283		

$F = 132.24 \sim F_{2,745}$ under H_0, and a p-value $(Pr > F)$ of $= .0001$

Why *df=a-1*?
Pretend the other factor wasn't measured. A sample of *a* X-bars from the distribution of X-bar.

The test

(2) Test for the presence of column effects: H_0: all $\beta_j = 0$ vs. H_1: at least one $\beta_j \neq 0$
This is a test for the effect of sex on SBP level after controlling for the effect of dietary group.

The null hypothesis for B: $H_B: \beta_1 = \beta_2 = \cdots = \beta_b = 0$

Under the null:

$$SS(B)/\sigma^2 \sim \chi^2(b-1)$$

$$F_B = \frac{SS(B)/(b-1)}{SS(E)/[ab(c-1)]} \sim F(b\text{-}1,\ ab(c\text{-}1))$$

Table 12.14 **Mean SBP by dietary group and sex**

Dietary group		Sex: Male	Sex: Female
SV strict vegetarians	Mean	109.9	102.6
LV lactovegetarians	Mean	115.5	105.2
NOR normals	Mean	128.3	119.6

SOURCE	DF	SS	F VALUE	PR > F
SEX	1	13056.29786994	66.65	0.0001
ERROR	745	145934.76850283		

$F = 66.65 \sim F_{1,745}$ under H_0, $p = .0001$

Why *df=b-1*?
Pretend the other factor wasn't measured. A sample of b X-bars from the distribution of X-bar.

The test

(3) Test for the presence of interaction effects: H_0: all $\gamma_{ij} = 0$ vs. H_1: at least one $\gamma_{ij} \neq 0$. This is a test of whether or not there is a differential effect of dietary group between males and females. For example, dietary group may have an effect on SBP only among men.

The null hypothesis for AB interactions: $H_{AB}: \gamma_{ij} = 0,\ i = 1, 2, \ldots, a;\ j = 1, 2, \ldots, b,$

Under the null:

$$\mathrm{SS(AB)}/\sigma^2 \sim \chi^2((a-1)(b-1))$$

$$F_{AB} = \frac{\mathrm{SS(AB)}/[(a-1)(b-1)]}{\mathrm{SS(E)}/[ab(c-1)]} \sim F(\,(a\text{-}1)(b\text{-}1),\ ab(c\text{-}1)\,)$$

Why *df=(a-1)(b-1)*?

$\sum \gamma_{ij} = 0$ for all j, $\sum \gamma_{ij} = 0$ for all i

A total of *ab* measurements.

The number of constraints in terms of γ_{ij} is *a+b-1*, hence losing *a+b-1* df.

The test

An Example—A Study of Iron Retention An experiment was performed to determine whether two forms of iron (Fe^{2+} and Fe^{3+}) are retained differently. (If one form of iron were retained especially well, it would be the better dietary supplement.) The investigators divided 108 mice randomly into 6 groups of 18 each; 3 groups were given Fe^{2+} in three different concentrations, 10.2, 1.2, and .3 millimolar, and 3 groups were given Fe^{3+} at the same three concentrations.

Fe^{3+}			Fe^{2+}		
10.2	1.2	.3	10.2	1.2	.3
.71	2.20	2.25	2.20	4.04	2.71
1.66	2.93	3.93	2.69	4.16	5.43
2.01	3.08	5.08	3.54	4.42	6.38
2.16	3.49	5.82	3.75	4.93	6.38
2.42	4.11	5.84	3.83	5.49	8.32
2.42	4.95	6.89	4.08	5.77	9.04
2.56	5.16	8.50	4.27	5.86	9.56
2.60	5.54	8.56	4.53	6.28	10.01
3.31	5.68	9.44	5.32	6.97	10.08
3.64	6.25	10.52	6.18	7.06	10.62
3.74	7.25	13.46	6.22	7.78	13.80
3.74	7.90	13.57	6.33	9.23	15.99
4.39	8.85	14.76	6.97	9.34	17.90
4.50	11.96	16.41	6.97	9.91	18.25
5.07	15.54	16.96	7.52	13.46	19.32
5.26	15.89	17.56	8.36	18.4	19.87
8.15	18.3	22.82	11.65	23.89	21.60
8.24	18.59	29.13	12.45	26.39	22.25

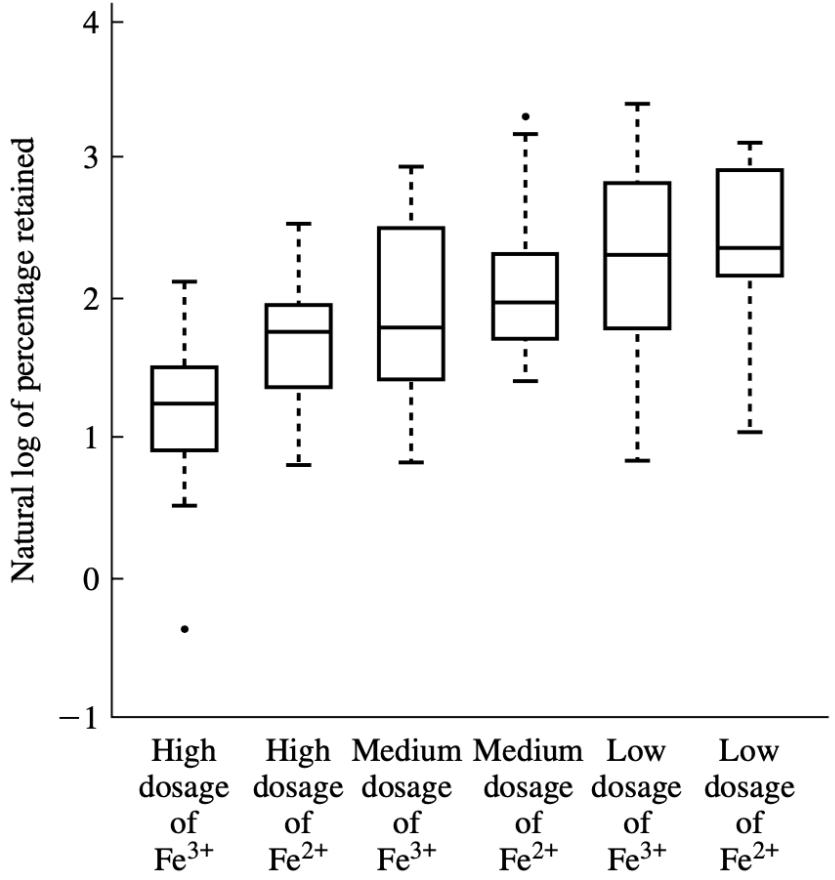


FIGURE **12.5** Boxplots of log data on iron retention.

The test

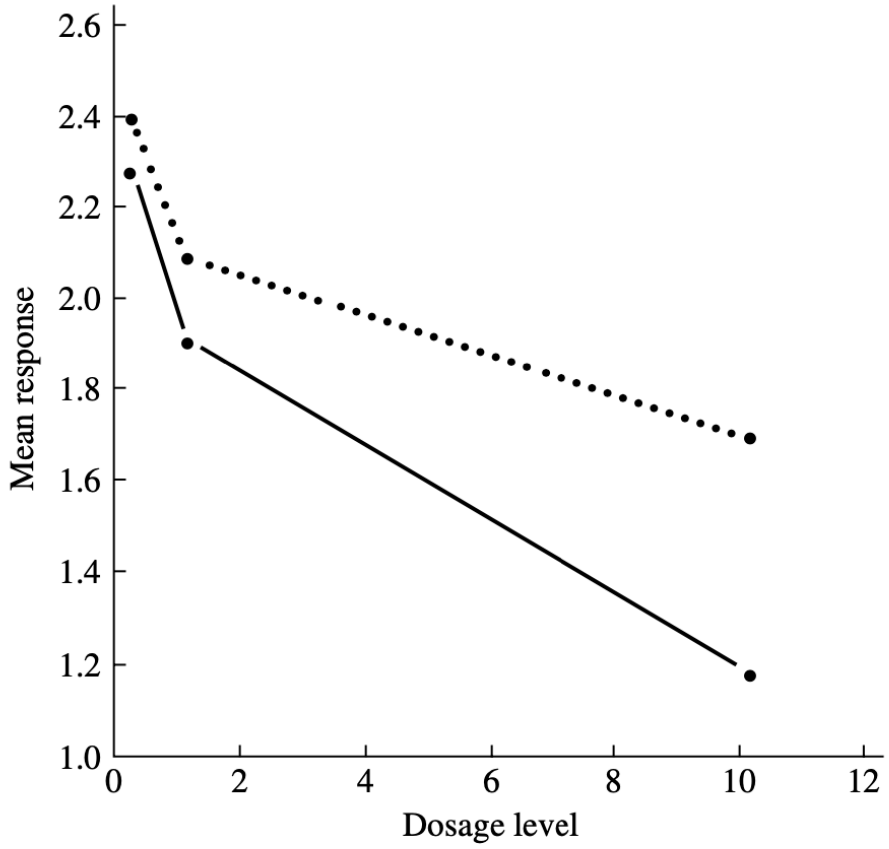


FIGURE **12.8** Plot of cell means of log data versus dosage level. The dashed line corresponds to Fe^{2+} and the solid line to Fe^{3+}.

Analysis of Variance Table

Source	df	SS	MS	F
Iron form	1	2.074	2.074	5.99
Dosage	2	15.588	7.794	22.53
Interaction	2	.810	.405	1.17
Error	102	35.296	.346	
Total	107	53.768		

$$H_A: \alpha_1 = \alpha_2 = 0$$

$$F = \frac{SS(Iron\ form)/2}{SS(E)/102} = 5.99$$

p-value =1-pf(5.99,1,102)= 0.016

The test

Analysis of Variance Table

Source	df	SS	MS	F
Iron form	1	2.074	2.074	5.99
Dosage	2	15.588	7.794	22.53
Interaction	2	.810	.405	1.17
Error	102	35.296	.346	
Total	107	53.768		

$$H_B: \beta_1 = \beta_2 = \beta_3 = 0$$

$$F = \frac{SS(Doage)/2}{SS(E)/102} = 22.53$$

p-value =1-pf(1.17,2,102) is approximately 0

The test

Analysis of Variance Table

Source	df	SS	MS	F
Iron form	1	2.074	2.074	5.99
Dosage	2	15.588	7.794	22.53
Interaction	2	.810	.405	1.17
Error	102	35.296	.346	
Total	107	53.768		

$$H_{AB}: \gamma_{11} = \gamma_{12} = \cdots = \gamma_{32} = 0$$

$$F = \frac{SS(Interaction)/2}{SS(E)/102} = 1.17$$

p-value =1-pf(1.17,2,102)= 0.31

What happens when there is one observation per cell?

The interaction term can no longer be included –

It allows each cell to have unconstrained mean value.

There will be perfect fit and SS(E)=0.

$$X_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$$

$$\overline{X}_{i\cdot} = \frac{1}{b}\sum_{j=1}^{b} X_{ij}$$

$$\overline{X}_{\cdot j} = \frac{1}{a}\sum_{i=1}^{a} X_{ij}$$

$$\overline{X}_{\cdot\cdot} = \frac{1}{ab}\sum_{i=1}^{a}\sum_{j=1}^{b} X_{ij}$$

j

i

Table 9.4-2 Gas mileage data

	Gasoline				
Car	1	2	3	4	$\overline{X}_{i\cdot}$
1	26	28	31	31	29
2	24	25	28	27	26
3	25	25	28	26	26
$\overline{X}_{\cdot j}$	25	26	29	28	27

What happens when there is one observation per cell?

$$SS(TO) = SS(A) + SS(B) + SS(E)$$

$$\sum_{i=1}^{a}\sum_{j=1}^{b}(X_{ij} - \overline{X}_{..})^2 = b\sum_{i=1}^{a}(\overline{X}_{i.} - \overline{X}_{..})^2 + a\sum_{j=1}^{b}(\overline{X}_{.j} - \overline{X}_{..})^2 + \sum_{i=1}^{a}\sum_{j=1}^{b}(X_{ij} - \overline{X}_{i.} - \overline{X}_{.j} + \overline{X}_{..})^2$$

Table 9.4-1 Two-way ANOVA table, one observation per cell

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F
Factor A (row)	$SS(A)$	$a-1$	$MS(A) = \frac{SS(A)}{a-1}$	$\frac{MS(A)}{MS(E)}$
Factor B (column)	$SS(B)$	$b-1$	$MS(B) = \frac{SS(B)}{b-1}$	$\frac{MS(B)}{MS(E)}$
Error	$SS(E)$	$(a-1)(b-1)$	$MS(E) = \frac{SS(E)}{(a-1)(b-1)}$	
Total	$SS(TO)$	$ab-1$		

What happens when there is one observation per cell?

Testing row effect:

The null hypothesis: $H_A: \alpha_1 = \alpha_2 = \cdots = \alpha_a = 0$

$$F_A = \frac{\mathrm{SS(A)}/(a-1)}{\mathrm{SS(E)}/[(a-1)(b-1)]} \quad \sim \quad F\ [a-1, (a-1)(b-1)]$$

Testing Column effect:

The null hypothesis: $H_B: \beta_1 = \beta_2 = \cdots = \beta_b = 0$

$$F_B = \frac{\mathrm{SS(B)}/(b-1)}{\mathrm{SS(E)}/[(a-1)(b-1)]} \quad \sim \quad F\ [b-1, (a-1)(b-1)]$$

What happens when there is one observation per cell?

Can we expect the same mileage for each of these four types of gasoline?

$$H_B: \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$$

$$\text{SS(B)} = 3[(25-27)^2 + (26-27)^2 + (29-27)^2 + (28-27)^2] = 30$$

$$\text{SS(E)} = (26-29-25+27)^2 + (24-26-25+27)^2 + \cdots + (26-26-28+27)^2 = 4.$$

$$\frac{\text{SS(B)}/(4-1)}{\text{SS(E)}/[(3-1)(4-1)]} = 15 > 9.78 = F_{0.01}(3,6)$$

H_B is rejected. The gasolines seem to give different performances.

Table 9.4-2 Gas mileage data

	Gasoline				
Car	1	2	3	4	$\overline{X}_{i\cdot}$
1	26	28	31	31	29
2	24	25	28	27	26
3	25	25	28	26	26
$\overline{X}_{\cdot j}$	25	26	29	28	27

What happens when there is one observation per cell?

Table 9.4-2 Gas mileage data

	Gasoline				
Car	1	2	3	4	$\overline{X}_{i\cdot}$
1	26	28	31	31	29
2	24	25	28	27	26
3	25	25	28	26	26
$\overline{X}_{\cdot j}$	25	26	29	28	27

Table 9.4-3 ANOVA table for gas mileage data

Source	Sum of Squares (SS)	Degrees of Freedom	Mean Square (MS)	F	p-value
Row (A)	24	2	12	18	0.003
Column (B)	30	3	10	15	0.003
Error	4	6	2/3		
Total	58	11			